

Reviewer Pack — Minimal Local Coherence and Frege Proof Depth Inequality

Entry 0de1ac382431 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Coherence and Frege Proof Depth Inequality

Entry ID: 0de1ac382431 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-08 11:18:56 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Topology (Local Systems) **Field B** (complexity object): Boolean Satisfiability (Frege Proof Complexity)

Statement:

For every boolean formula φ with n variables, the local coherence of its associated simplicial complex χ_φ is linearly correlated with its Frege proof depth $d(\varphi)$, such that $\text{local_coherence}(\chi_\varphi) = \Theta(d(\varphi))$.

Rationale (proposer's reasoning):

Algebraic topology has been successfully applied to study properties of Boolean formulas through the lens of simplicial complexes. Local coherence, a measure from algebraic topology, could potentially capture intrinsic geometric aspects of these complex structures, which may be related to the complexity of Frege proofs. A deeper understanding of such relationships may lead to new insights into proof complexity.

Taxonomy category: Category Theory (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): b239a2662b9dc603

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the Pearson correlation coefficient between local coherence and Frege proof depth exceeds 0.7 for all formulas φ with $n \leq 40$, across at least 30 different seeds, and no seed produces a correlation below 0.5.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - algebraic topology AND local systems AND Frege proof complexity - Frege proof depth AND simplicial complex AND algebraic topology - Boolean satisfiability AND minimal local coherence AND proof complexity

Top relevant hits considered: - [http://arxiv.org/abs/1111.0731v2] On Hawaiian Groups of Some Topological Spaces - [http://arxiv.org/abs/1907.00871v6] Cardinal-indexed classifying spaces for families of subgroups of any topological group - [http://arxiv.org/abs/1706.09157v2] Topological complexity of the work map - [http://arxiv.org/abs/1802.03129v3] Rank Selection and Depth Conditions for Balanced Simplicial Complexes - [http://arxiv.org/abs/2506.12921v2] Shortest Paths in a Weighted Simplicial Complex - [http://arxiv.org/abs/2309.11304v3] A new quantum computational set-up for algebraic topology via simplicial sets - [http://arxiv.org/abs/1109.2049v1] Structure-Based Local Search Heuristics for Circuit-Level Boolean Satisfiability - [http://arxiv.org/abs/1102.1242v2] A refinement of Stone duality to skew Boolean algebras

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_formula(n):
        if n == 1:
            return random.choice(['True', 'False'])
        else:
            op = random.choice(['&', '|'])
            subformulas = [generate_formula(random.randint(1, n-1)) for _ in range(2)]
            return f'({subformulas[0]} {op} {subformulas[1]})'

    def evaluate_formula(phi, assignment):
        if phi == 'True':
            return True
        elif phi == 'False':
            return False
        else:
            var, op, subphi = phi.split()
            val_var = assignment[var]
            val_subphi = evaluate_formula(subphi, assignment)
```

```

        if op == '&':
            return val_var and val_subphi
        elif op == '|':
            return val_var or val_subphi

def compute_simplicial_complex(phi):
    n = phi.count('(') + 1
    simplicial_complex = []
    stack = []
    for i, char in enumerate(phi):
        if char == '(':
            stack.append(i)
        elif char == ')':
            start = stack.pop()
            simplicial_complex.append(phi[start:i+1])
    return simplicial_complex

def compute_local_coherence(simplicial_complex):
    # Simplified version of local coherence calculation
    return len(simplicial_complex) / n

def compute_frege_proof_depth(phi):
    if phi == 'True' or phi == 'False':
        return 1
    else:
        var, op, subphi = phi.split()
        return 2 + max(compute_frege_proof_depth(subphi), compute_frege_proof_depth(subphi))

n_values = [5, 10, 15, 20, 30, 40]
results = []

for n in n_values:
    for _ in range(5):
        phi = generate_formula(n)
        assignment = {var: random.choice([True, False]) for var in set(phi) if var.isalpha()}
        simplicial_complex = compute_simplicial_complex(phi)
        local_coherence = compute_local_coherence(simplicial_complex)
        frege_depth = compute_frege_proof_depth(phi)
        results.append((local_coherence, frege_depth))

mean_local_coherence = sum(x for x, _ in results) / len(results)
mean_frege_depth = sum(y for _, y in results) / len(results)
correlation_coefficient = (n - 1) * sum((x - mean_local_coherence) * (y - mean_frege_depth) for x, y in results) / (math.sqrt(sum((x - mean_local_coherence)**2 for x, _ in results)) * math.sqrt(sum((y - mean_frege_depth)**2 for _, y in results)))

return {
    "metric_name": "correlation_coefficient",
    "metric_value": correlation_coefficient,
    "instances_tested": len(results),
    "n_max": max(n_values),
    "conjecture_holds": correlation_coefficient > 0.7,
    "counterexample": "" if correlation_coefficient >= 0.5 else "correlation_below_0.5"
}

if __name__ == "__main__":

```

```

import sys
seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

results = []
for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction=1.0")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fraction}")
else:
    first_failing_seed = next(i for i, r in enumerate(results) if not r["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"correlation_below_0.5\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca240dd3.py", line 99, in <module>
    result = run_trial(seed)
              ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca240dd3.py", line 75, in run_trial
    frege_depth = compute_frege_proof_depth(phi)
                  ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca240dd3.py", line 63, in compute_frege
    var, op, subphi = phi.split()
    ~~~~~
ValueError: too many values to unpack (expected 3)

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which means it was unable to compute the required metrics for the conjecture. | next: Investigate and fix the crash in the test code to ensure it can complete the computation of local coherence and Frege proof depth.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14215
2	preregistration	ollama_remote	glm4:latest	0	0	12491
3	novelty	ollama_remote	glm4:latest	0	0	8305
4	novelty	ollama_remote	glm4:latest	0	0	15382
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19300
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12516
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17136
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11962
9	judge	ollama_remote	glm4:latest	0	0	26934

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 138241 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/0de1ac382431.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/0de1ac382431.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/0de1ac382431.tar.gz` (if generated)